

Part 1 What does energy look like?

The core activities in this unit total about 22.5 hours of lesson time. The Content Descriptions (Australian Curriculum: Science) for each activity are listed here as curriculum codes and described fully in each lesson plan.

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
1.1	Spot the energy	ACSSU155 ACSIS148	Engage & Explore	Students explore a range of phenomena and identify different types and stores of energy.	Core
			Classroom & Digital		
			Medium		
1.2	Energy to burn	ACSSU155 ACSIS148 ACSHE136	Engage & Evaluate	Students watch a video of a rocket launch and identify some of the energy forms and transfers occurring. They will recall words associated with energy from Activity 1.1 and will learn two terms associated with mechanical energy: kinetic and gravitational potential energies.	Core
			Short		
			Medium		
1.3	Forms of energy	ACSSU155 ACSIS148	Explore, Explain & Evaluate	Students will label a magazine picture, identifying all energy related phenomena.	Core
			Classroom		
			Short		

Part 2 Why do things move?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
2.1	Throw the ball	ACSSU155 ACSIS145 ACSIS148	Explore & Explain	Students will observe someone throwing a ball as high in the air as possible. They will discuss the forces involved while the ball is being thrown and during its flight. They will also identify forms of mechanical energy involved; gravitational potential and kinetic energies, and how a transformation between these two forms of energy allows for the conservation of mechanical energy.	Core
			Classroom & Digital		
			Short		
2.2	Energy skate park	ACSSU155 ACSIS234	Explain & Elaborate	Students will explore the Energy skate park learning object and apply their understanding of kinetic and gravitational potential energies to explain the skater's motion.	Optional
			Digital		
			Medium		
2.3	Energy in, energy out	ACSSU155 ACSIS148	Explore & Explain	Students will explore the concept of work through an investigation of simple levers and pulley systems.	Core
			Hands-on		
			Long		
2.4	Heat from friction	ACSSU155 ACSIS144 ACSIS145 ACSIS234 ACSIS148	Explore	Students will conduct a simple experiment exploring the link between friction and the heating of water.	Core
			Hands-on		
			Medium		
2.5	Starting fires	ACSSU155 ACSHE226 ACSIS148	Explore & Explain	Students learn about the use of fire through human history and indigenous perspectives of the use of fire for human survival.	Optional
			Hands-on		
			Long		
2.6	Putting it together	ACSSU155 ACSHE226 ACSHE136	Elaborate & Evaluate	Students will evaluate their understanding through class discussion and completion of a quiz.	Core
			Classroom		
			Medium		

Part 3 What is heat? What is temperature?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
3.1	What is heat?	ACSSU155 ACSHE226	Engage & Explain	Students are introduced to a range of phenomena that relate to the concept of heat. They then watch a video about Rumford's discovery that heat was a form of energy that could be generated through friction. Students explore their own ideas and formulate their own hypotheses as they progress through the activity.	Core
			Digital		
			Medium		
3.2	Keeping your drink warm	ACSSU155 ACSIS139 ACSIS140 ACSIS141 ACSIS234 ACSIS148	Engage & Explain	Students explore different ways of keeping a hot drink warm, and relate these to conduction & convection.	Core
			Hands-on		
			Medium		
3.3	Water is hard to heat	ACSSU155 ACSIS139 ACSIS144 ACSIS145 ACSIS234	Explore, Explain & Elaborate	Students explore the relationships between heat, temperature, volume and heat capacity, particularly of water.	Core
			Hands-on		
			Long		
3.4	Feeling hot and cold	ACSSU155 ACSHE136 ACSIS139 ACSIS144 ACSIS234	Explore & Explain	Students will learn about the importance of conduction in how hot or cold surfaces feel.	Core
			Hands-on & Digital		
			Medium		
3.5	Heating water with light	ACSSU155 ACSIS144 ACSIS145 ACSIS146 ACSIS234 ACSIS148	Explain	Student will conduct a laboratory experiment to determine whether the colour of a surface affects the heating effect of light.	Core
			Digital		
			Medium		
			Hands-on & Digital		
			Long		

Part 3 What is heat? What is temperature? (continued)

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
3.6	Solar energy	ACSSU155 ACSIS140 ACSIS141 ACSIS144 ACSIS145 ACSIS146 ACSIS234	Explore, Explain, Elaborate & Evaluate	Students design and test a solar oven for heating water. They will follow a basic plan to design a solar collector that will collect and focus as much light on a container of water to maximise heating.	Core
3.7	What is sunlight?	ACSSU155 ACSIS144	Explore & Explain	Students will use a light box to visualise the spectrum of light. Notebook questions draw together their ideas about light and colour.	Core
			Hands-on & Digital		
			Medium		
3.8	Cyclones	ACSSU155	Explain & Elaborate	Students explore digital resources and apply knowledge to explain how the sun's energy drives a cyclone.	Optional
			Digital		
			Medium		

Part 4 What is electricity?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
4.1	Making a globe light up	ACSSU155 ACSIS140 ACSIS144	Engage & Explore	Students discover the fundamentals of making an electric circuit work.	Core
			Hands-on		
			Medium		
4.2	Showing electric circuits	ACSSU155 ACSIS144	Explore	Students will design their own symbols to represent components in an electric circuit.	Core
			Classroom		
			Short		
4.3	Heating with electricity	ACSSU155 ACSIS140 ACSIS141 ACSIS145	Explore & Explain	Students construct an electric water heater using high resistance wire.	Optional
			Hands-on		
			Medium		
4.4	Build a motor	ACSSU155 ACSHE 134 ACSIS140 ACSIS145	Explore & Elaborate	Students will follow instructions to build a model motor.	Core
			Hands-on & Digital		
			Medium		
4.5	Build a generator	ACSSU155 ACSHE 134 ACSHE 135 ACSIS144 ACSIS145 ACSIS234	Explore & Elaborate	Students build a simple generator using basic equipment.	Core
			Hands-on		
			Short		
4.6	The lemon battery	ACSSU155 ACSIS144 ACSIS145	Explore & Explain	Students explore the best way to make an LED glow using a lemon battery.	Optional
			Hands-on		
			Medium		

Part 5 Why do we eat?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
5.1	Food as fuel	ACSSU155 ACSIS140 ACSIS144 ACSIS145	Explore & Explain	Students will conduct a simplified calorimetric experiment to measure the energy content of a nut	Core
			Hands-on		
			Medium		
5.2	Energy in food	ACSSU155	Explore	Students investigate the energy content of a range of foods and relate this to daily needs. They judge the quality of different foods.	Core
			Classroom & Digital		
			Medium		
5.3	Who eats what?	ACSSU155 ACSIS148	Explain, Elaborate & Evaluate	Students use the Classroom Antarctica website to build an Antarctic food web. They will complete a simple exercise linking food chains and food webs to the transfer of chemical energy through ecosystems.	Core
			Classroom		
			Short		
5.4	Rates of photosynthesis	ACSSU155 ACSIS139 ACSIS140 ACSIS141 ACSIS144 ACSIS145 ACSIS148	Explain, Elaborate & Evaluate	Students carry out a digital activity and conduct an experiment to investigate one of the factors affecting photosynthesis.	Core
			Hands-on & Digital		
			Long		

Part 6 How do we energise our cities, towns and homes?

Activity No	Activity Name	Content Descriptions	Lesson type	Activity Description	
6.1	Where does our energy come from?	ACSSU155 ACSHE 134 ACSHE 135 ACSHE136	Elaborate & Evaluate	Students investigate, using on-line and local resources, where their electricity supply comes from and how it is generated.	Core
			Digital		
			Medium		
6.2	Energy systems	ACSSU155 ACSIS148	Elaborate	Students examine several systems involving multiple energy transfers and use a Sankey diagram to represent and analyse each (e.g. energy transfer in a car and domestic household).	Core
			Digital		
			Medium		
6.3	Conserving energy	ACSSU155	Explore, Explain, Elaborate & Evaluate	Within the context of their own location, students explore ways of minimising heat transfer in a house to save energy used in heating or cooling. Students will design and evaluate an energy efficient house that minimises the movement of heat into or out of the house.	Optional
			Digital & Classroom		
			Medium		
6.4	Honda ad	ACSSU155 ACSHE136 ACSIS145 ACSIS148	Engage, Explain & Evaluate	Students watch the link to the UK Honda Accord commercial and identify: - energy forms and transfers - energy stores in the system.	Core
			Digital		
			Medium		
6.5	Sample test		Evaluate	A sample summative test (editable version) and marking scheme are available to teachers by email request from <i>Science by Doing</i> (sbd@science.org.au). In your email request please include your school (or university) and contact details. Note - <i>Science by Doing</i> provides sample assessment items and whilst every effort has been made, the security of these items cannot be guaranteed. <i>Science by Doing</i> encourages teachers to modify the items to suit individual teaching programs.	Optional
			Classroom		
			Long		

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